

NT-C-023 | SQL | Course Knowledge Profile

Identity, purpose, prerequisites, scope and boundaries

Category	Data & Databases
Target learners	Learners seeking database, reporting and analytics foundations.
Prerequisites	Basic computer use; programming is not required.
Approved duration	6-8 weeks / 60-90 guided hours.
Final outcome	Design and query relational databases accurately, using joins, analytics, transactions and performance basics.
Assessment profile	Data/AI

Approved tools / platforms

PostgreSQL, MySQL or another approved relational database; SQL client and version-controlled scripts.

Knowledge boundaries

Include	Quality rule	Exclude / escalate
The eight approved modules, four sections, named tools, projects and mapped entry-level roles.	Require data provenance, cleaning decisions, train/test separation, suitable metrics, reproducibility and responsible interpretation.	Reject data leakage, invented metrics, unsupported causal claims and hidden manual corrections.
Current product/platform procedures only when version and date are known.	Use primary documentation and record the source version.	Do not invent current commands, prices, tax rates, policies or certification status.
Evidence-based career guidance.	Cite tests, projects, capstone, viva and verified resume evidence.	No guaranteed job, placement, salary, interview or employer claim.

LLM answer pattern: identify the course and version, answer from approved records, cite record IDs, distinguish verified facts from guidance, and state any missing evidence.

NT-C-023 | SQL | NT-C-023-S01 Deep Knowledge

Modules 1-2 | objectives, concepts, evidence and remediation

Section competency	Tables, keys, relationships and normalisation; SELECT, filters, sorting and expressions
Completion evidence	Schema exercise; Query set
Section weight	20% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 1 - Relational foundation

Objective: Tables, keys, relationships and normalisation

Observable evidence: Schema exercise

Concept	Approved knowledge statement	Application behaviour	Evidence
Tables	A defined competency within SQL. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when tables is appropriate; apply it in a SQL task; verify the result.	Schema exercise
keys	A defined competency within SQL. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when keys is appropriate; apply it in a SQL task; verify the result.	Schema exercise
relationships	A defined competency within SQL. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when relationships is appropriate; apply it in a SQL task; verify the result.	Schema exercise
normalisation	A defined competency within SQL. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when normalisation is appropriate; apply it in a SQL task; verify the result.	Schema exercise

Module 2 - Query basics

Objective: SELECT, filters, sorting and expressions

Observable evidence: Query set

Concept	Approved knowledge statement	Application behaviour	Evidence
SELECT	A defined competency within SQL. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when select is appropriate; apply it in a SQL task; verify the result.	Query set
filters	A defined competency within SQL. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when filters is appropriate; apply it in a SQL task; verify the result.	Query set
sorting	A defined competency within SQL. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when sorting is appropriate; apply it in a SQL task; verify the result.	Query set
expressions	A defined competency within SQL. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when expressions is appropriate; apply it in a SQL task; verify the result.	Query set

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Schema exercise; Query set, return to Modules 1-2, complete one guided practice and then use a new equivalent test form.

NT-C-023 | SQL | NT-C-023-S02 Deep Knowledge

Modules 3-4 | objectives, concepts, evidence and remediation

Section competency	Inner/outer joins and set operations; GROUP BY, HAVING and subqueries
Completion evidence	Multi-table queries; Summary reports
Section weight	25% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 3 - Joins

Objective: Inner/outer joins and set operations

Observable evidence: Multi-table queries

Concept	Approved knowledge statement	Application behaviour	Evidence
Inner/outer joins	A defined competency within SQL. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when inner/outer joins is appropriate; apply it in a SQL task; verify the result.	Multi-table queries
set operations	A defined competency within SQL. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when set operations is appropriate; apply it in a SQL task; verify the result.	Multi-table queries

Module 4 - Aggregation

Objective: GROUP BY, HAVING and subqueries

Observable evidence: Summary reports

Concept	Approved knowledge statement	Application behaviour	Evidence
GROUP BY	A defined competency within SQL. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when group by is appropriate; apply it in a SQL task; verify the result.	Summary reports
HAVING	A defined competency within SQL. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when having is appropriate; apply it in a SQL task; verify the result.	Summary reports
subqueries	A defined competency within SQL. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when subqueries is appropriate; apply it in a SQL task; verify the result.	Summary reports

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Multi-table queries; Summary reports, return to Modules 3-4, complete one guided practice and then use a new equivalent test form.

NT-C-023 | SQL | NT-C-023-S03 Deep Knowledge

Modules 5-6 | objectives, concepts, evidence and remediation

Section competency	DDL, constraints and data modification; CTEs, window functions and date analysis
Completion evidence	Controlled schema; Analytics queries
Section weight	25% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 5 - Data definition

Objective: DDL, constraints and data modification

Observable evidence: Controlled schema

Concept	Approved knowledge statement	Application behaviour	Evidence
DDL	A defined competency within SQL. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when ddl is appropriate; apply it in a SQL task; verify the result.	Controlled schema
constraints	A defined competency within SQL. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when constraints is appropriate; apply it in a SQL task; verify the result.	Controlled schema
data modification	A defined competency within SQL. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when data modification is appropriate; apply it in a SQL task; verify the result.	Controlled schema

Module 6 - Analytical SQL

Objective: CTEs, window functions and date analysis

Observable evidence: Analytics queries

Concept	Approved knowledge statement	Application behaviour	Evidence
CTEs	A defined competency within SQL. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when ctes is appropriate; apply it in a SQL task; verify the result.	Analytics queries
window functions	A defined competency within SQL. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when window functions is appropriate; apply it in a SQL task; verify the result.	Analytics queries
date analysis	A defined competency within SQL. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when date analysis is appropriate; apply it in a SQL task; verify the result.	Analytics queries

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Controlled schema; Analytics queries, return to Modules 5-6, complete one guided practice and then use a new equivalent test form.

NT-C-023 | SQL | NT-C-023-S04 Deep Knowledge

Modules 7-8 | objectives, concepts, evidence and remediation

Section competency	Transactions, views, permissions and indexes; Business schema, queries and optimisation notes
Completion evidence	Reliable database lab; SQL portfolio
Section weight	30% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 7 - Reliability

Objective: Transactions, views, permissions and indexes

Observable evidence: Reliable database lab

Concept	Approved knowledge statement	Application behaviour	Evidence
Transactions	A defined competency within SQL. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when transactions is appropriate; apply it in a SQL task; verify the result.	Reliable database lab
views	A defined competency within SQL. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when views is appropriate; apply it in a SQL task; verify the result.	Reliable database lab
permissions	A defined competency within SQL. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when permissions is appropriate; apply it in a SQL task; verify the result.	Reliable database lab
indexes	A defined competency within SQL. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when indexes is appropriate; apply it in a SQL task; verify the result.	Reliable database lab

Module 8 - Capstone

Objective: Business schema, queries and optimisation notes

Observable evidence: SQL portfolio

Concept	Approved knowledge statement	Application behaviour	Evidence
Business schema	A defined competency within SQL. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when business schema is appropriate; apply it in a SQL task; verify the result.	SQL portfolio
queries	A defined competency within SQL. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when queries is appropriate; apply it in a SQL task; verify the result.	SQL portfolio
optimisation notes	A defined competency within SQL. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when optimisation notes is appropriate; apply it in a SQL task; verify the result.	SQL portfolio

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Reliable database lab; SQL portfolio, return to Modules 7-8, complete one guided practice and then use a new equivalent test form.

NT-C-023 | SQL | Skill Ontology & Dependency Map

Atomic skills, learning order and evidence strength

Skill ID	Competency	Depends on	Evidence	Type
NT-C-023-SK01	Relational foundation	Prerequisite foundation	Schema exercise	Critical
NT-C-023-SK02	Query basics	M01 plus earlier foundations	Query set	Critical
NT-C-023-SK03	Joins	M02 plus earlier foundations	Multi-table queries	Supporting
NT-C-023-SK04	Aggregation	M03 plus earlier foundations	Summary reports	Supporting
NT-C-023-SK05	Data definition	M04 plus earlier foundations	Controlled schema	Supporting
NT-C-023-SK06	Analytical SQL	M05 plus earlier foundations	Analytics queries	Critical
NT-C-023-SK07	Reliability	M06 plus earlier foundations	Reliable database lab	Supporting
NT-C-023-SK08	Capstone	M07 plus earlier foundations	SQL portfolio	Critical

Evidence-strength hierarchy

Strength	Evidence source	Use
High	Trainer-observed practical, viva, working source/project files and reproducible output.	May support verified mastery when rubric threshold is met.
Medium	Scored mock tests, reviewed assignments and documented portfolio artifacts.	Supports mastery with corroboration.
Low	Resume claim, self-report, attendance or video completion without demonstration.	May trigger verification; cannot alone support Apply Now.

CareerTours must not treat attendance, time spent or content opened as skill mastery.

NT-C-023 | SQL | Assessment Knowledge & Sample Items

Non-live examples for LLM training and evaluation

Sample ID	Type	Question	Approved answer / rubric anchor
NT-C-023-S01-EX01	Evidence/application	What evidence best demonstrates the combined competency of Relational foundation and Query basics?	The learner should provide Schema exercise; Query set and explain how the work applies Tables, keys, relationships and normalisation; SELECT, filters, sorting and expressions. Recall alone is insufficient.
NT-C-023-S01-EX02	Scenario/remediation	A learner reports completing Query basics but cannot produce Query set. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-023-S01, request Query set, and allow a new equivalent test only after remediation.
NT-C-023-S02-EX01	Evidence/application	What evidence best demonstrates the combined competency of Joins and Aggregation?	The learner should provide Multi-table queries; Summary reports and explain how the work applies Inner/outer joins and set operations; GROUP BY, HAVING and subqueries. Recall alone is insufficient.
NT-C-023-S02-EX02	Scenario/remediation	A learner reports completing Aggregation but cannot produce Summary reports. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-023-S02, request Summary reports, and allow a new equivalent test only after remediation.
NT-C-023-S03-EX01	Evidence/application	What evidence best demonstrates the combined competency of Data definition and Analytical SQL?	The learner should provide Controlled schema; Analytics queries and explain how the work applies DDL, constraints and data modification; CTEs, window functions and date analysis. Recall alone is insufficient.
NT-C-023-S03-EX02	Scenario/remediation	A learner reports completing Analytical SQL but cannot produce Analytics queries. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-023-S03, request Analytics queries, and allow a new equivalent test only after remediation.
NT-C-023-S04-EX01	Evidence/application	What evidence best demonstrates the combined competency of Reliability and Capstone?	The learner should provide Reliable database lab; SQL portfolio and explain how the work applies Transactions, views, permissions and indexes; Business schema, queries and optimisation notes. Recall alone is insufficient.
NT-C-023-S04-EX02	Scenario/remediation	A learner reports completing Capstone but cannot produce SQL portfolio. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-023-S04, request SQL portfolio, and allow a new equivalent test only after remediation.

Live bank rule: Generate at least three times the live quantity for each section using the approved Data/AI blueprint. Human-review every scored item before initial production release.

NT-C-023 | SQL | Labs, Projects & Scoring Rubrics

Practical proof of learning and ownership

Level	Approved project	Skills evidenced	Required deliverables
Mini 1	Student database	Schema and CRUD	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Mini 2	Sales report suite	Joins, aggregates and windows	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Capstone	Operations analytics database	Design, queries and performance	Brief; working/source file or reproducible environment; output; validation notes; reflection.

Standard project rubric

Dimension	Weight	What earns credit
Problem framing	15%	Clear objective, user/business need, assumptions and success criteria.
Technical/design accuracy	25%	Correct methods, appropriate tools and sound decisions.
Completeness	15%	Required features, assets, data or workflow are present.
Testing and validation	15%	Important normal, edge, quality or failure cases are checked.
Evidence and reproducibility	15%	Source/working files, setup, inputs, outputs and version information are available.
Documentation	5%	Concise structure, instructions and limitations.
Viva / ownership	10%	Learner can explain decisions, demonstrate work and respond to a change request.

Require data provenance, cleaning decisions, train/test separation, suitable metrics, reproducibility and responsible interpretation.

NT-C-023 | SQL | Career Roles, Evidence & Gap Actions

Explainable top-three occupation matching

Rank	Occupation	Default weighted skills	Minimum evidence	Gap action
1	SQL Developer Trainee	SQL 34% [critical]; schema 33% [critical]; transactions 33%	Script portfolio	If a critical skill is below 60 or script portfolio is missing, return Improve First or Needs Evidence with the linked module/project.
2	Database Support Trainee	Queries 34% [critical]; permissions 33% [critical]; diagnostics 33%	Database lab	If a critical skill is below 60 or database lab is missing, return Improve First or Needs Evidence with the linked module/project.
3	Reporting/Data Analyst - Junior	Analytical SQL 50% [critical]; KPIs 50% [critical]	Business report queries	If a critical skill is below 60 or business report queries is missing, return Improve First or Needs Evidence with the linked module/project.

Course readiness	45% section mocks + 20% mini projects + 25% capstone + 10% viva.
Role readiness	45% role-relevant mastery + 25% projects/capstone + 20% aligned mocks + 10% verified resume evidence.
Evidence confidence	Show separately; Apply Now requires at least 70.
Resume extraction	Use only relevant skills, projects, education, experience and portfolio evidence; minimise personal data before external model processing.

A student may be Apply Now for one mapped occupation and Improve First for another. Return independent evidence and gaps for each role.

NT-C-023 | SQL | FAQs, Misconceptions & LLM Response Pack

Course-specific retrieval and response patterns

Approved FAQ	Answer
Who is SQL for?	Learners seeking database, reporting and analytics foundations.
What should I know before starting?	Basic computer use; programming is not required.
How long is the approved learning journey?	6-8 weeks / 60-90 guided hours.
What will I be able to demonstrate?	Design and query relational databases accurately, using joins, analytics, transactions and performance basics.
Does completion guarantee a job?	No. Completion and role readiness are evidence-based guidance; employment, placement, salary and employer selection are not guaranteed.
What if I fail a section?	CareerTours identifies the failed skill tags and module, assigns targeted practice, and permits an equivalent retest after remediation.

Misconception	Approved correction
Attendance proves mastery	False - require tests and observable practical/project evidence.
One total score proves every role fit	False - calculate each occupation using its weighted skills and evidence.
The LLM can add missing details	False - retrieve approved records or state that evidence is insufficient.
Course completion guarantees employment	False - never claim guaranteed job, placement, salary or interview.
SQL knowledge never changes	False - version-sensitive tools, laws, taxes, security and platform procedures require review.

Retrieval metadata

```
course_code=NT-C-023; corpus_version=3.0;
record_types=course_profile,module,section,skill,assessment_sample,project,occupation_mapping,faq,misconception;
approval_status=approved; effective_date=2026-08-12
```